

 Pakistan National Accreditation Council	NON-CONFORMITY REPORT FOR ASSESSMENT VISIT		F-02/06 Issue Date: 11/11/19 Rev No: 06

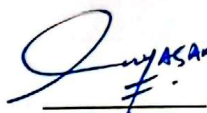
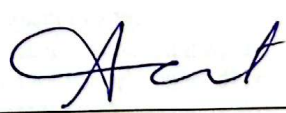
ACTIVITY:	☒ Assessment ☐ Surveillance		NCR No: 4
	☐ Re-Assessment ☐ Scope extension		Accr./Appl.No:
	☐ Other Please specify:		Non-conformity category: Very serious ☐ Essential ☒
			Time limit for correction: 6 weeks
ORGANIZATION:	Quality control Lab vital Agri. Nutrients (Pvt) Ltd.		Mr. Hammyat Hussain
Department:	Labs		
Team Leader:	Mr. Atique Ur Rehman	Mngt. Rep.	
Assessor	Dr. Muhammad Aftab		
Expert			
DESCRIPTION: The Lab has developed a procedure related to metrological traceability (QSP-LSP-05) The reference standards used are valid and traceable to their primary standards and stored in refrigerator but monitoring the temperature through un calibrated thermo hygrometer (NC). 10-01-2025 _____ Date Signature Assessor/Expert Signature (Management Representative)			Requirement of: ISO/IEC 17025 ☒ ISO 15189 ☐ ISO/IEC 17021 ☐ ISO/IEC 17020 ☐ Others:
			Ref. Organization's doc.
CORRECTIVE ACTIONS: (Note: It is mandatory to attach Root Cause Analysis) Corrective action for shortcoming on NC has been taken by the calibration of thermo hygrometer for monitoring the temperature. Action are documented in the amendment no: <u>CA-04-02-2025</u> <u>18-02-25</u> Date Signature (Management Representative)			☐ It is not necessary to attach documentation
REASON FOR CLOSING: (To be filled in by the Team Leader.) ☐ The non-conformity is closed based on satisfactory documentation from the organization ☐ The non-conformity is closed based on recommendation from the technical assessor ☐ Implementation of the corrective actions will be followed up at the next visit Remarks (if any) ----- ----- ----- The non-conformity is closed: _____ Date Signature (Team Leader.)			

	CORRECTIVE ACTION REQUEST	Doc. No.	QCL-FRM-9.01
		Issue Date	01-10-2024
		Rev#	01

Initiated by: Masab Rasheed	Department: QC Lab	CAR No. CA-04-02-2025
Designation: AQCM	CAR Initiated Date: 03-02-2025	NC No. 04

CAR Initiated due to (Please tick the relevant appropriate box below)

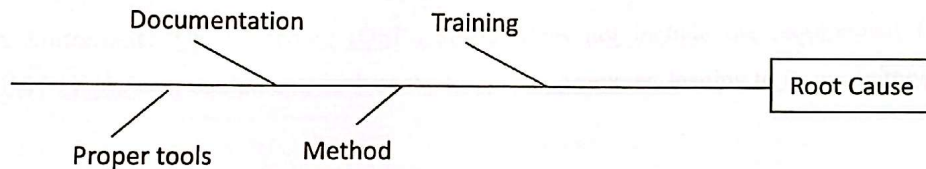
- ☒ PNAC Assessment
 ☐ Complaint
 ☐ Accident/Incident
 ☐ Internal Audit
☐ Suggestion/Improvement
 ☐ Technical Fault
 ☐ Others

A. Description (to be filled by Initiator) The Lab has developed a procedure related to metrological traceability (QSP-LSP-05) The reference standards used are valid and traceable to their primary standards and stored in refrigerator but monitoring the temperature through un calibrated thermo hygrometer (NC).		
 Signatures		
Significance/Priority Level	☒ Urgent ☐ Normal ☐ Non-significance	
☒ Accepted ☐ Rejected	Assigned to: Masab Rasheed Date: 03-02-2025	
* If accepted then Manager QC to mark it to concerned person for Root Cause Analysis.		
B. Root Cause Analysis (where required) The procedure (QSP-LSP-05) does not include the requirement for calibration of auxiliary monitoring equipment, such as the thermo-hygrometer, leading to its use without calibration.		
C. Proposed Action	Target Date:	18-02-2025
Corrective action for shortcomings on NC has been taken by the calibration of thermo hygrometer for monitoring the temperature.		
 Name or Signatures (Person who carried out Root Cause Analysis and Proposed Corrective Action)		
D. Verification by Manager QC for implementation & Effectiveness		
Is Proposed Action Implemented?	☒ YES ☐ NO	Is it Effective? ☒ YES ☐ NO
Remarks (If any):		
CAR closing date: <u>18-02-25</u>		Manager QC Sign: 

	ROOT CAUSE ANALYSIS FORM	Doc. No.	QCL-FRM-9.04
		Issue Date	01-10-2024
		Rev#	01

Non-Conformance Description			
The Lab has developed a procedure related to metrological traceability (QSP-LSP-05) The reference standards used are valid and traceable to their primary standards and stored in refrigerator but monitoring the temperature through uncalibrated thermo hygrometer (NC).			
RCA Team Members	Himayat Hussain (QCM), Masab Rasheed (AQCM)		
RCA Team Lead / Head	Himayat Hussain (QCM)		
RCA Start Date	03-02-2025	Submission Date	18-02-2025
Methodology / Tools Used			
Cause & Effect Diagram	☐	5WHY analysis	☐
		3W2H / 5W2H	☒

Cause Effect Diagram



3W2H / 5W2H	Phenomenon of Problem / Question	Explanation / Answer
WHY	Why was the temperature monitoring of the refrigerator using an uncalibrated thermo-hygrometer?	There was no calibration schedule or tracking mechanism for the thermo-hygrometer.
WHY	Why was there no calibration schedule for the thermo-hygrometer?	Calibration of auxiliary monitoring devices (like thermo-hygrometers) was not considered a priority or overlooked during procedure development.
WHY	Why was calibration of the thermo-hygrometer overlooked?	The focus was on ensuring traceability and validity of primary reference standards, but monitoring tools (support equipment) were not included in the calibration scope.
WHY	Why was monitoring tools not included in the calibration scope?	There was a gap in the procedure (QSP-LSP-05) that did not specify the need for calibration of monitoring equipment like thermo-hygrometers.
WHY	Why was there a gap in the procedure (QSP-LSP-05)?	The risk assessment during the procedure's development did not account for the potential impact of uncalibrated monitoring equipment on traceability.

	ROOT CAUSE ANALYSIS FORM	Doc. No.	QCL-FRM-9.04
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How	How to address this specific NC (Corrective Action)?	Identify all monitoring equipment (like the thermo-hygrometer) used in the lab and send them for calibration to an accredited body. Update the refrigerator monitoring records with data from the calibrated equipment.
How	How to prevent this NC from recurring (Preventive Action)?	• Revise the procedure (QSP-LSP-05) to include requirements for periodic calibration of all monitoring equipment. • Conduct training sessions for relevant staff on the importance of calibrating auxiliary equipment. • Implement a system to track calibration schedules for all lab equipment, including monitoring devices.
Root Cause: The procedure (QSP-LSP-05) does not include the requirement for calibration of auxiliary monitoring equipment, such as the thermo-hygrometer, leading to its use without calibration.		

RCA Statement: The procedure (QSP-LSP-05) does not include the requirement for calibration of auxiliary monitoring equipment, such as the thermo-hygrometer, leading to its use without calibration.

Reviewed by: 

Evaluated by: 



GOVERNMENT OF PAKISTAN
Ministry of Science & Technology
Pakistan Council of Scientific & Industrial Research
Laboratories Complex, Lahore

CALIBRATION CERTIFICATE

Certificate # APCIC/TL (374)

Issue Date: 17 February 2025

Customer Details:

Customer Name: Vital Agri Nutrients (Pvt.) Ltd.,
Customer Address: 49 km Multan Road Near Pakistan CNG Ganga Road, Kasur.

Request Ref. #: Nil Dated: 7th February, 2025

ILO #/Date: 776/17.02.25

Details of Unit Under Calibration (UUC):

Thermo Hygrometer

CONTROLLED COPY
NO. _____

Serial # / Type: Nil / Digital I.D. Code: QC-INS-024

Make / Model: China / HTC-2 Ranges / Parameters: Given Below

Calibration Date: 13-Feb-2025 Calibration Due(If Required): 12-Feb-2026

Conditions of UUC ☒ Found ok upon receipt ☐ Manual received alongwith UUC

Environmental Conditions during measurement:

Temperature: 19.4 °C $\pm$ 0.7 °C Humidity: 62.8 %rh $\pm$ 2.8 %rh

Calibration Method / Procedure (s): APCIC/TCP/03

Traceability Statement:

The measurements made by Calibration Labs, APCIC, PCSIR realize the physical units of measurement (SI) through National Institute of Metrology, Pakistan NMIP (USNO, NIST USA, A member of BIPM & APMP). Following Reference Equipment are used for measurement(s) which are maintained & controlled at APC&I Centre, PCSIR Laboratories Complex, Lahore

Reference Equipment:

Equipment	ID #	Make / Model	Certificate #	Traceability
Thermo Hygrometer	Cal-Lab-TI22	TEKLINX	CC # 16882	NMIP
Humidity Chamber	Cal-Lab-THG-01	TABAI ESPEC Corp.	APCIC/TL(1829)	NMIP

CALIBRATION RESULTS

Cal. Range / Points(s): i) 15 to 30 °C ii) 49 to 80 % rh Resolution: i) 0.1 °C ii) 1 % rh

Temperature (Inner Sensor):

Sr. #	Nominal reading on the UUC X (°C)	Mean Value measured by Reference Standard A (°C)	Error X-A (°C)	Expanded Uncertainty $\pm$ (°C)
1	15.0	14.8	0.2	0.27
2	20.0	20.1	-0.1	0.27
3	25.0	25.1	-0.1	0.27
4	30.0	30.0	0.0	0.27

Terms and conditions over leaf

Prepared by:

17/02/25
MUHAMMAD MUNIR
Technical Officer
PCSIR Laboratories Complex
Lahore

DR. ZEESHAN ALI
Head, APC&I Centre
PCSIR LABS COMPLEX LAHORE

Counter signature

99-02-25
Director General

Object No. 042-99230574, PABX No.042-99230688-95, Ext. 216-17, 363, Fax: 042-99230705
Email: pcsirlabhr@gmail.com; website: www.pcsir-lab.gov.pk



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CALIBRATION CERTIFICATE

Certificate # APCIC/TL

(374)

ILO # 776/17.01.25

Issue Date: 17th February, 2025

Thermohygrometer

CONTROLLED COPY
NO.

Humidity:

Sr. #	Nominal reading on the UUC X (%rh)	Mean Value measured by Reference Standard A (%rh)	Error X-A (%rh)	Expanded Uncertainty ± (%rh)
1	49	39.7	9.3	1.9
2	60	49.8	10.2	1.9
3	70	59.5	10.5	1.9
4	80	70.3	9.7	1.8

The reported expanded uncertainty is based on combined uncertainty multiplied by a coverage factor $k=2$, providing a coverage probability of approximately 95%.

Remarks / Statement of Conformity:

- ☒ Mean Value based on Six readings ☐ Adjustment / Offset adjustment (if any) ☐ On-site Calibration
- ☐ Stabilization / Warm up Time (if any) ☐ Control limits provided by customer ☐ Revised Cal. Certificate
- ☒ Customer signed T&Cs of Cal Labs before calibration activity ☐ * Non Accredited Activity
- ☒ Statement of conformity not given in this certificate as process control limits were not conveyed by the customer.
- ☐ The equipment complies with the process control limits used as specifications given by customer at the measured points (S), with due allowance having been made for uncertainty of the measurements. The decision rule deployed to issuing this statement of conformity is taking into account the measurement uncertainty with zero guard band.
- ☐ The measured values along with their respective uncertainty bands fall within the process control limits and that's the unit under calibration conforms to the specifications (process control limits given by customer).
- ☐ The measured values along with their respective uncertainty bands do not fall within the process control limits and thus the unit under calibration does not conform to the specifications (process control limits given by customer).

♦.....♦ End of Calibration Results ♦.....♦

Terms and conditions over leaf

Prepared by:

17/02/25
MUHAMMAD MUNIR
Senior Laboratory Charge
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Lahore

DR. ZEESHAN ALI
Head, Applied Physics Division
PCSI Laboratories Complex
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18-02-25
Director General

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